Land Sharing or Land Sparing? Implications for Farmland Birds of Conservation Concern in France

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Abstract

Purpose: Farmland bird populations have experienced widespread declines across Europe, raising urgent questions about how agricultural landscapes should be structured to reconcile biodiversity conservation and food production. Land sparing and land sharing represent two contrasting strategies, yet empirical evidence remains mixed.
Objectives: We aim to quantify responses of bird communities along a land sharing–sparing gradient and identify which landscape management strategies best support different ecological groups.
Methods: Using the French Breeding Bird Survey, we analyzed 810 locations monitored between 2001–2024. A land sharing–sparing gradient was quantified using forest area, agricultural heterogeneity, and landscape diversity indices. Bird counts were modeled with negative binomial spatio-temporal Bayesian models

(`INLA`), incorporating climate, spatial random fields, and temporal autocorrelation.
Results: Forest specialists responded positively to greater forest area, supporting land sparing. Generalists benefited from higher agricultural heterogeneity, consistent with land sharing. Unexpectedly, farmland specialists did not benefit from increased heterogeneity, and conservation-priority farmland species showed negative associations with natural elements.
Conclusion: No single land-use strategy maximizes abundance across all bird groups. Land sparing favors forest specialists and conservation-priority species, while land sharing benefits generalists. Context-dependent landscape planning and custommed agricultural practices are essential.

Keywords: Land sharing and land sparing, farmlands birds, INLA, conservation

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1 Introduction

With the current global biodiversity decline (Burns et al. 2021; Inger et al. 2015), land use allocation is a major challenge for biodiversity conservation. Agricultural landscapes are central to this issue, as they cover a large proportion of European territory and harbour a substantial part of biodiversity.

Several studies show that farmland bird populations are declining in Europe (Donald et al. 2001; Bowler et al. 2019). This decline is mainly explained by the intensification of agricultural practices (Rigal et al. 2023; Genty et al. 2026) and pesticide use (Monnet et al. 2026; Perrot et al. 2025). Heldbjerg et al. (2018) also shows that ground-nesting birds are in significant decline in Denmark whereas birds nesting elsewhere are less impacted. This raises the question of how land should be allocated to maximize biodiversity? Two main land-use strategies are commonly discussed (Fischer et al. 2014; Grass et al. 2019; Fahrig 2020, 2013): land sparing and land sharing. The **land sparing** approach separates highly intensive agricultural production from conservation areas in order to minimize trade-offs between food production and biodiversity preservation. In such landscapes, agricultural areas are typically characterized by large, homogeneous patches and intensive farming practices, which often result in low biodiversity. In contrast, biodiversity is concentrated in relatively large protected areas such as national nature reserves or forests. By contrast, the **land sharing** approach aims to support both food security and biodiversity by promoting their coexistence within the same landscapes. These landscapes are generally composed of many small agricultural fields interspersed with trees, hedgerows, or semi-natural elements.

As a result, they tend to be more heterogeneous, with higher connectivity between habitat patches, and farming practices are usually less intensive.

Today, there is no consensus on how to allocate land use to maximize biodiversity and agricultural output. Indeed, on the one hand Phalan et al. (2011) and Edwards et al. (2015) show that land sparing is better to conserve biodiversity in Ghana, India and Colombia. On the other hand, Riva and Fahrig (2022) show that small patches contain higher diversity. Moreover, Lamb et al. (2019) show that land sparing would more likely only benefit forest birds and would disadvantage farmland birds in the United Kingdom. Finch et al. (2019) proposes a solution based on three compartments enabling some large forest patches to conserve woodland birds, some low-intensity agriculture to conserve farmland birds and some big fields to ensure food security.

These previous results suggest that the effect of land sharing or land sparing on bird abundance depends on both the landscape and species' needs. These needs correspond to the ecological niches of the species which may be narrow for specialist species and broader for generalists (Devictor et al. 2010).

To resolve this uncertainty in the French agricultural landscapes, which covers more than half the country with highly diverse farming practices (Nagy and Soós 2025), we need robust monitoring approaches. Birds serve as ideal indicators because they are easy to sample, cover a wide range of life-history traits, are present in all habitats and trophic levels (Padoa-Schioppa et al. 2006; Karen Aghababyan 2024), and are sensitive to environmental stressors (Canterbury et al. 2000; Gregory et al. 2003; Rigal et al. 2023).

Moreover, citizen science facilitates bird data collection because many volunteers are skilled at bird identification. Hence, thanks to thousands of skilled French bird-watchers we gathered an extensive database from a breeding bird survey called the French Breeding Bird Surveys (Jiguet et al. 2012; Fontaine et al. 2021).

As indicators of land sharing and land sparing are complex, we need to take into account many variables that structure the landscape and bird populations. Moreover, to account for the spatio-temporal structure, a Bayesian framework and inference with `INLA` (Rue et al. 2009) appears to be particularly suitable because it efficiently handles the hierarchical structure of spatio-temporal data with multiple covariates.

In this study, we use spatio-temporal Bayesian modeling to disentangle the effects of land sharing and land sparing scenario on bird populations, with a focus on farmland and forest birds with a focus on species of conservation concern. We formulated the following hypotheses:

1. Land sharing is expected to benefit generalist and farmland species. In land-sharing landscapes, the higher diversity of habitat patches may provide a wider range of nesting opportunities for generalist species. For farmland species, lower agricultural intensity under land sharing is predicted to promote breeding success and survival.
2. Land sparing is expected to benefit forest species, particularly those of conservation concern. Forest specialists are generally sensitive to human disturbance; therefore, larger contiguous areas allocated to biodiversity conservation under land sparing are predicted to enhance their abundance.

2 Materials and Methods

2.1 Data

2.1.1 Bird abundance data

The data are provided by the French Breeding Bird Surveys (Jiguet et al. 2012; Fontaine et al. 2021). It aims to survey the common French breeding bird populations over time and space in France. This is a participatory program where thousands of skilled volunteers will count birds. These volunteers get randomly assigned a 2 km by 2 km square, next to the place they live, by the French National Museum of Natural History. Once this square is assigned, the volunteer needs to choose ten evenly distributed points inside the square. Then, each year the volunteer has to visit the square twice during breeding season to record bird abundance. A visit of the square consists of staying 5 minutes at each point and recording every seen or heard bird (Jiguet et al. 2012).

In order to have clean data, we removed every square that has not been observed in the right times and dates. We also removed squares that do not have ten points in it (around 2% of the data). Then we chose to aggregate data to the scale of the square (sum of all points) to feasibly carry out statistical analysis made by `INLA` and remove detectability bias (Nabias et al. 2024).

To have enough temporal depth in our analysis we only kept time series of at least five years. As our goal is to study the effect of different kinds of agricultural practices on bird population, we removed all squares that have three or more point classified as urban by the observer or at least 25% of urban land use in the square.

It left us with a total of 810 squares that have been observed at least five years in a row. We used the French Breeding Bird Surveys classification (Fontaine et al. 2021) of bird to categorize birds with their main types of habitat adding two species to agricultural birds: *Passer montanus* (Barlow et al. 2020) and *Burhinus oedicnemus* (Hume and Kirwan 2020).

To disentangle the effects of land sharing and land sparing on French bird populations, we divided birds into groups of habitat specialization (agricultural, forest and generalist birds) given by Fontaine et al. (2021) and Julliard et al. (2006). Then, we used the French IUCN list (UICN France et al. 2016) to classify the species in two categories: least concerned and birds with conservation concern. In the French IUCN list (UICN France et al. 2016) they mention birds for which the category changes between 2008 and 2016. As our dataset goes from 2001 to 2024, we considered that a species is with conservation concern if it is not "least concerned" in 2008 or 2016.

Maps of distribution of birds' observation by group are shown in Supplementary material Section 1 (Figure SI-1 for generalist species, Figure SI-2 for forest species, Figure SI-3 for agricultural species, Figure SI-4 for forest species with conservation concern and Figure SI-5 for agricultural species with conservation concern).

2.1.2 Environmental data

Climate data is taken from WordClim (Harris et al. 2020). For each square and year we retrieved minimum and maximum temperatures and total amount of precipitation of the current and last spring (March to June) as well as minimum and maximum of last summer (July and August). Then we added the latitude, recentered to be positive when in the North and negative otherwise. This variable also serves through interactions with the location and climate variables, allowing us to check if our model works well by finding the same results as in Thepault et al. (2025).

We used CORINE land cover (European Environment Agency 2019) to retrieve land cover variables. This database contains all land uses in France in a raster format for years 2000, 2006, 2012 and 2018. The different kind of land uses are described in 44 groups (Section 5 of Supplementary material). For each year, we retrieved in each square the surface of agricultural (all CORINE codes starting by 2) and water areas (all CORINE codes starting by 5), forest (all CORINE codes starting by 31), natural (all CORINE codes starting by 32 or 33), water (all CORINE codes starting by 4 or 5) and artificial lands (all CORINE codes starting by 1). This provides a general assessment of the use of each square. We did a linear interpolation of the data when it was not constant between two years of data.

2.1.3 Land sharing and sparing indicators

To create indicator of land sharing and land sparing we took the intersection of CORINE patches and a disk of radius $10km$ centered in the middle of each square. This allowed us to obtain, for each square, the number of patches intersecting the disk, their area within the disk and their type. As the area of forest around the square is a great indicator of land sparing we retrieved the total area of forest patches intersecting a disk of 10 km radius around the center of the square (without any distinction between different forest types).

In order to have an indicator of the heterogeneity of agricultural practices, we computed a Shannon index (Shannon 1948) on agricultural land uses around the squares where the "species" is the kind of patch and the abundance the number of each kind of patch. Shannon's indices will give us an idea of the diversity of agricultural land uses around each square. If this index is high then there is a high diversity in land uses, saying we are more in a land sharing scenario. Otherwise, it means that we are in a monocultural scenario meaning that we are in a land sparing scenario. An example of the land use around a square is given in Figure 1.

The last two variables we added are the forest area and agriculture with significant amount of nature in a radius of 10 kilometers around the point of interest. A higher forest area implies a land sparing scenario whereas a higher amount of agriculture with significant amount of nature implies a land sharing scenario.

2.1.4 Correlation study and Principal Component Analysis (PCA) on the land cover variables

The land use of each square consists of six highly correlated variables as they all sum up to the area of the square (4 square kilometers). We then use a PCA to summarize these variables onto fewer variables that are not correlated. After computations, we can summarize into three component explaining 80.8% of the variance of the data (Figure SI-7c). The axes are represented in Figures SI-7a and SI-7b. The first axis separates

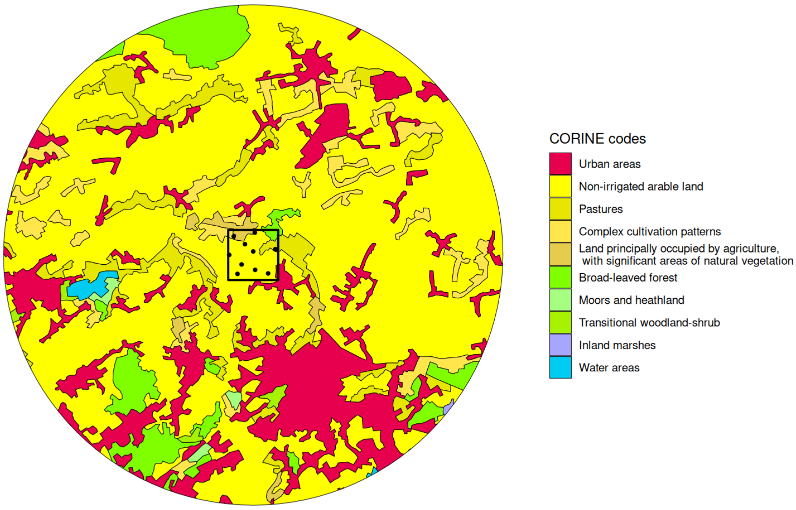

Fig. 1 Example of the land use in a 10 km radius disk around a square (black line). Each color represents a different kind of land use following the table in Supplementary material Section 5 (colors used are the official CORINE colors). Shannon's indices are calculated only with agricultural patches (codes starting with a two). In this case there are 68 agricultural patches spread in four categories. The black dots represent ten points of samples.

the agricultural squares (negative) from the natural and forest squares (positive). The second axis separates urban (and water) squares (positive) with the others. The third axis separates closed squares (forest and cities) from the open squares (flat lands). We call these axes PCA1, PCA2 and PCA3.

2.2 Statistical approach

2.2.1 Model

Bird abundance may vary due to multiple drivers such as land-use change, climate variability, agricultural practices, and interactions with other species. To quantify these effects, we modelled bird counts using a negative binomial regression model, which accounts for overdispersion in the data due to the lack of observation as well as for real bird distribution. Formally, the expected abundance at location s and year t is defined as:

$$\log \lambda(s,t) = \beta_0 + \sum_k \beta_k X_k(s,t) + \sum_{i,j} \beta_{i,j} X_i(s,t) X_j(s,t) + w_{s,t},$$

where $X_k(s,t)$ denote environmental covariates (e.g. climate, land cover), and $w_{s,t}$ represents residual spatio-temporal variability. Interaction terms, $X_i(s,t)X_j(s,t)$, allow us to capture non-additive effects between environmental drivers. In our case, we used the separable case for the spatio-temporal variability:

$$w_{t,s} = w_s + w_t,$$

where w_t is an autoregressive model and w_s is a Matérn random field. For the temporal effect, there is a correlation between years, modelled in the following way:

$$w_t = \phi w_{t-1} + \epsilon_t,$$
$$\epsilon_t \sim \mathcal{N}(0, \tau^{-1}),$$

where ϕ is the auto-correlation parameter and τ is the precision parameter. For the spatial Matérn random field, the covariance is as following (as in Krainski et al. (2018)):

$$\mathrm{Cov}(w_s, w_{s'}) = \sigma^2 \frac{1}{2^{\nu-1}\Gamma(\nu)} (\kappa \|s - s'\|)^{\nu} K_{\nu}(\kappa \|s - s'\|);$$

where: σ^2 is the marginal variance of the process; ν is the smoothness parameter (usually set to 1); κ is a scaling parameter (linked to the range ρ of the process by: $\rho = \sqrt{8\nu}/\kappa$); and K_ν is the modified Bessel function of the second kind. The range ρ of the process gives information on the spatial correlation range between points.

2.2.2 Inference

We relied on a Bayesian framework for inference. Instead of using the traditional Monte Carlo Markov Chain algorithm to get posterior marginals, we used the Integrated Nested Laplace Approximation (`INLA`) developed by Rue et al. (2009). It provides fast and accurate deterministic approximations of posterior distributions for latent Gaussian models.

In particular, it computes posterior marginals for latent effects (regression coefficients) as well as for hyperparameters (variance of spatial and temporal random effects). This approach makes it possible to fit computationally demanding spatio-temporal models efficiently.

Prior choice. As it is a Bayesian framework, we have to choose priors. They are summarized in Table 1. For the spatial field, we chose to use penalized complexity (PC) priors (Simpson et al. 2017) because it helps in choosing less complex model by penalizing the complexity of the latent Gaussian random field.

The spatial random field needs two parameters: the range, giving information about the spatial correlation range between two points and the variance of the distance of interaction σ (in the Matérn field). We chose to set the interaction range at 100 km because birds return to the same nesting sites (Jiguet et al. 2012; Fontaine et al. 2021; Barbet-Massin et al. 2012). For the variance we chose $\sigma = 1$ as a weakly informative prior.

All the other priors are default `INLA` priors that are weakly informative and are summarized in Table 1.

Mesh choice. The first thing to do is to choose a mesh for the spatial random field. The mesh is automatically constructed by `INLA`, but we need to choose three parameters: the inner edge, the outer edge and the cutoff value. To avoid any boundary effects in the analysis, the mesh should be extended further away from the spatial domain (Lindgren et al. 2011). The inner edge sets the maximum length of an edge inside the spatial domain. In Dambly et al. (2023) they suggest to use inner edge as

Table 1 Summary of prior distributions used in the INLA spatio-temporal models. PC stands for Penalized Complexity

Model component	Prior distribution	Interpretation / Notes
Spatial random field	PC on range and σ	$\Pr(\text{range} < 100\,000) = 0.99$, $\Pr(\sigma > 1) = 0.01$
SPDE smoothness	Fixed to $\alpha = 2$	Corresponds to a Matérn field with moderate smoothness ($\nu = 1$)
Temporal random effect	PC priors on ϕ and τ	Shrinks toward $\phi = 0$ (no auto-correlation) and small variance
Fixed effects	$\mathcal{N}(0, 10^6)$	Weakly informative / nearly flat prior
Negative binomial dispersion	$\log(\theta) \sim \text{Gamma}(1, 0.00005)$	Weak prior on overdispersion parameter

1/5 the prior range, we assumed an interaction range of approximately 100 km, so we took 20 km. The outer edge sets the maximum length of an edge outside the spatial domain. In Righetto et al. (2020) they show that the outer range does not make so much difference and shall be bigger than the prior range, we chose 200 km. The cutoff value sets the minimal distance between two points in the mesh. As we cannot have observations that are less than 2 km away we chose the cut-off to be 2 km.

In Krainski et al. (2018) they mention that the mesh needs to be even in size and angle of the triangles of the mesh. It is the case with the values we chose (see Figure SI-6).

2.2.3 Model selection

We have five groups of species of interest : woodland species, farmland species, generalist species, woodland species with conservation concern and farmland species with conservation concern. We want to test the following set of variables: latitude, minimum temperature of last spring and last summer, maximum temperature of last spring and last summer, total amount of rain of last spring, the first three variables from the PCA on the land cover of each square (PCA1, PCA2, PCA3 from Section 2.1.4), the Shannon index on agricultural practices (Section 2.1.3), total surface of forest in a radius of 10 kilometers around the center of the square and the surface of farmland with significant amount of nature in a radius of 10 kilometers around the center of the square. To these variables, we added interactions between on the one hand maximum temperatures (of last spring and summer) and on the other hand latitude, surface of forest 10 kilometers around, surface of agriculture with significant amount of nature 10 kilometers around and Shannon index. This yielded a total of 12 variables and 8 interactions to test.

Testing possible model combinations for each group of species would be computationally intensive (around 10 seconds per model with more than 4000 models by group). We thus chose to perform stepwise selection to find the covariates that explain best the data, as in Illian et al. (2013). The criterion we used to select is the Widely Applicable Information Criterion (WAIC) as for example in Carson and Mills Flemming (2014). This criterion decreases when the fit of the model is better and penalizes the number of covariates. The WAIC gives a criterion of the goodness of fit of the model. However, it is a criterion that we can only use to compare models that are fitted to the exact same data and mesh parametrization (Dambly et al. 2023).

To test a wide range of models, we performed stepwise selection in three ways (Burnham and Anderson 2004): forward, backward and both. The forward procedure starts with the model with only random effects and then adds variables one by one. At each step, it selects the best variable to add to the model, according to the decrease of the WAIC score. When the two variables of an interaction are added, the corresponding interaction variable is added. Then, the backward procedure starts with the full model (all covariates and interaction) and remove one by one the covariates or interaction term along the same principle. Note that a covariate cannot be removed if an interaction term between this variable and another one is still in the formula. Finally, the "both" procedure starts as the forward procedure. After choosing a new variable to add, we add a backward step: we try to remove one by one each variable added in previous steps. We keep the model with the lowest WAIC after this backward step.

3 Results

We performed model selection for our five categories of birds. Detailed results can be found in Supplementary Table SI-1 for agricultural birds, Supplementary Table SI-2 for forest birds, Supplementary Table SI-3 for generalist birds, Supplementary Table SI-4 for agricultural birds with conservation concern and Supplementary Table SI-5 for forest birds with conservation concern.

First, we observed that in each category PCA1 has been chosen and is significant. Recall that this variable represents farmland squares when negative and forest squares when positive (Supplementary Figure SI-7a). Then all variables have been chosen at least in one group of birds. Regarding interactions, three of them have never been chosen by the selection procedure: Shannon index with last spring maximum temperature, Shannon index with last summer maximum temperature, and latitude with last summer maximum temperature.

Then, Table 2 and 3 indicate that, for all groups of species, there is a strong temporal autocorrelation ($\phi \approx 0.9$) and that the range of the Matérn field is not bigger than 40 km, meaning that the probability that two points more than 40 km apart depend on each other is very low. This distance aligns with the knowledge about bird breeding ecology as the breeding site fidelity and the distance of natal dispersal is known to be around 14 km (with 16 km of standard deviation) (Barbet-Massin et al. 2012). We can also observe that the overdispersion parameter is relatively high (from 11.64 to 47.25 Table 2 and 3).

Fixed effects are shown in Figure 2 for agricultural birds (A), forest birds (B) and generalist birds (C) and in Figure 3 for agricultural (A) and forest (B) birds with conservation concern.

The models use two kinds of **control variables** (land use and climate). Firstly, the **land uses** are described by the three first PCA axes: PCA1 (negative for agricultural squares and positive for forest squares), PCA2 (positive for urban square) and PCA3 (negative for open squares and positive for closed squares) (see 2.1.4). PCA1 is significant (when the confidence interval does not include zero) for all group of birds, positive for forest birds (**0.44**, with Confidence Interval $[0.40, 0.49]$), forest birds with conservation concern (**0.78** $[0.67, 0.88]$) and generalist birds (0.03 $[0.01, 0.06]$), and negative for farmland birds (**-0.52** $[-0.58, -0.46]$) and farmland birds with conservation concern (**-0.62** $[-0.68, -0.55]$). PCA2 is selected and significant for forest birds (-0.07 $[-0.10, -0.04]$) and forest birds with conservation concern (0.08 $[0.01, 0.16]$). PCA3 is selected and significant for agricultural birds (-0.23 $[-0.27, -0.18]$) agricultural birds with conservation concern (-0.30 $[-0.35, -0.26]$) and generalist birds (0.03 $[0.01, 0.05]$). These results are in line with the usual hypotheses that the abundance of a group of birds increases in the habitat they are specialized in.

Then the other set of control variables was composed of **climate variables** and latitude. Last spring minimum temperature was selected and significant for forest birds (0.03 [0.01,0.04]). Last spring maximum temperature was selected and significant for forest birds with conservation concern (-0.08 [-0.14,-0.03]). It was also selected but not significant for forest birds, agricultural birds and agricultural birds with conservation concern. Last spring total amount of rain was selected and significant for agricultural birds with conservation concern (-0.02 [-0.04,-0.01]). Last summer minimum temperature was selected but not significant for forest birds with conservation concern. Last summer maximum temperature was selected but not significant for agricultural birds, forest birds, and forest birds with conservation concern. The interaction between last spring maximum temperature and latitude was selected and significant for forest birds with conservation concern (-0.05 [-0.08,-0.02]). Although some climate variables were selected and significant for some models, we observe that these have limited effects because of the very small coefficients. However, when climate variables were selected and significant, results are globally in line with those of (Thepault et al. 2025).

In our approach the **land sharing and sparing** gradient is described thanks to three variables: agriculture with significant amount of nature, Shannon index and forest area in a radius of 10 kilometers around.

For agricultural birds (Figure 2-A) agriculture with a significant amount of nature and forest area 10 kilometers around were selected but not significant (respectively -0.05 [-0.11,0.01] and -0.03 [-0.11,0.04]). Some interaction between these variables and climate variables were also selected but not significant.

For forest birds (Figure 2-B), two variables of interest were selected and significant: forest area and agriculture with a significant amount of nature 10 kilometers around (respectively 0.17 [0.11,0.23] and 0.13 [0.08, 0.18]). Interactions between overall forest area and last spring and last summer maximum temperature were selected and significant (respectively -0.1 [-0.02, -0.01] and 0.01 [0.01, 0.02]).

For generalist birds (Figure 2-C) the Shannon index on agricultural diversity was selected and significant (0.15 [0.12,0.17]).

These results on all species primarily indicate that a land sparing scenario is better for forest and agricultural species and a land sharing scenario is better for generalist species.

We next examined the responses of species of conservation concern (Figure 3). For agriculture birds with conservation concern (Figure 3-A), the only land sharing/sparing variable selected was agriculture with a significant amount of nature and

Model component		Agricultural birds	Forest birds	Generalist birds
Spatial Random field	ρ (m)	17435.26	18336.21	13071.93
	σ	0.79	0.72	0.39
Temporal model	ϕ	0.89	0.60	0.89
	τ	111.48	367.41	359.25
Negative binomial dispersion		11.64	47.25	42.30

Table 2 Mean values of the posterior distribution of hyperparameters

is significant (-0.11 [-0.18,0.05]). For forest birds with conservation concern all three variables have been selected but only the area of forest 10 kilometers around was significant (**0.5** [0.37,0.70]). These results on birds with conservation concern showed that land sharing is not the preferred solution in order to conserve these species.

Model component		Agricultural birds	Forest birds
Spatial Random field	ρ (m)	18480.75	36334.33
	σ	1.01	1.46
Temporal model	ϕ	0.92	0.92
	τ	129.18	24.74
Negative binomial dispersion		29.51	12.87

Table 3 Mean values of the posterior distribution of hyperparameters for birds with conservation concern

4 Discussion

In this study, we aimed to disentangle the impacts of land sharing and land sparing on French bird populations. Note that in a land sparing scenario, the goal is to divide landscape into a zone of high production agriculture and an area dedicated to biodiversity conservation whereas in a land sharing scenario, the landscape is shared between areas of food production and biodiversity conservation (with for example natural structures inside farmland areas).

First, our results showed that in the French bird community there is a diversity of response to the land sharing and land sparing gradient depending on habitat specialization (farmland, woodland, generalist) and conservation concern. Hence, land sharing seems to be beneficial for **generalist species**, as the abundance of this group increases if there is more diversity in agricultural practices. This result aligns with the main hypothesis because a land sharing scenario includes small and many kinds of habitats that offer a wide variety of nesting sites and resources that benefit species that do not have strict requirements in terms of resources such as habitat or food (Martin and Fahrig 2018). At the scale of our study (a 10 km buffer around 810 sites across France) land sparing seems to be the best scenario for all other species group with no conservation status. Indeed, for **forest species** our results showed that land sparing scenario is better, because the abundance of this group increases with a greater

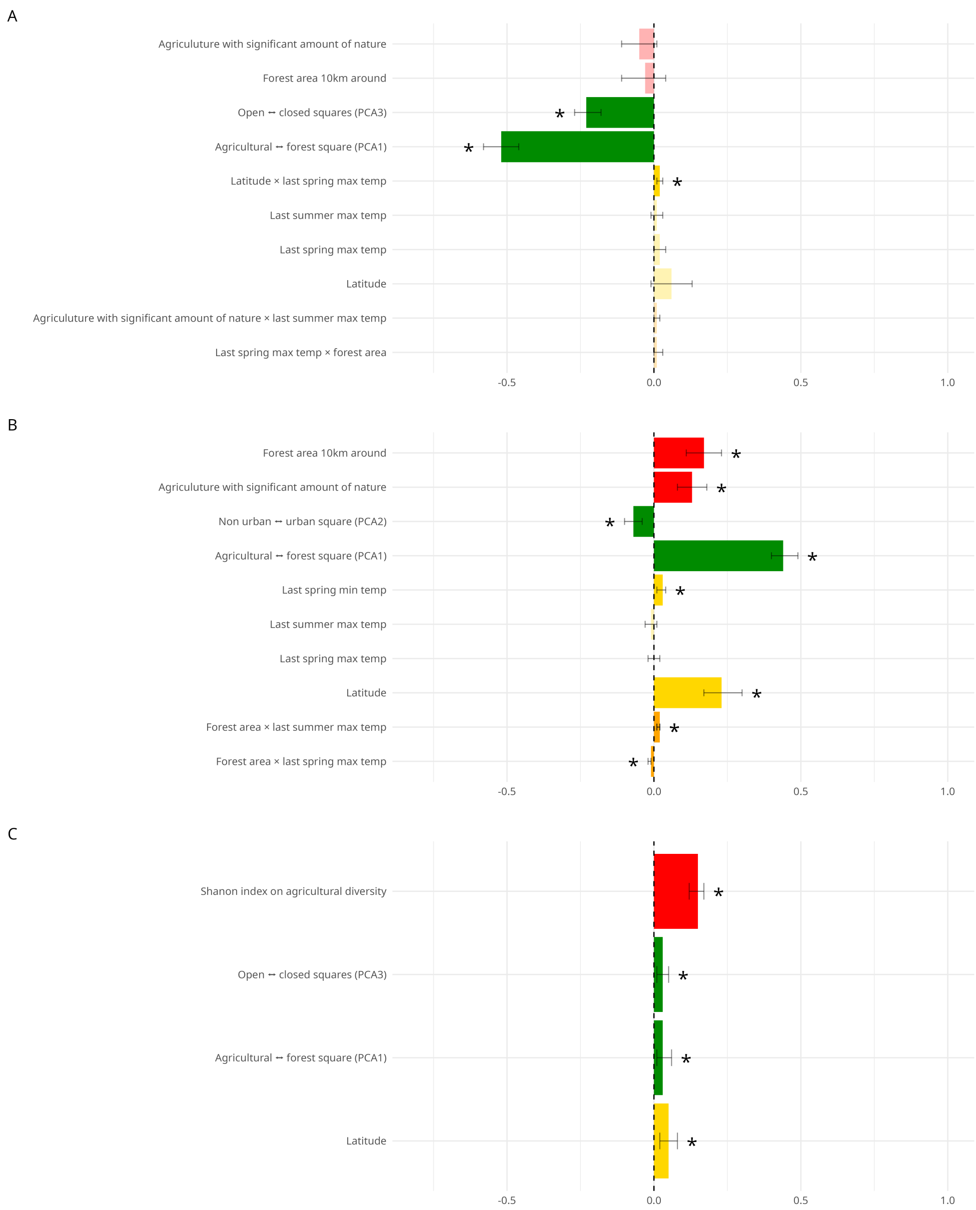

Fig. 2 Fixed effects of the best model for farmland species (A), woodland species (B) and generalist species (C). The variables highlighted in red are the land sharing and land sparing variables, the ones in green are about the land use of the squares, the one in yellow represent climate variables and the orange one are interactions between land sharing and sparing variables and climate. Significant variables are represented in plain color and with a star whereas non-significant variables are transparent.

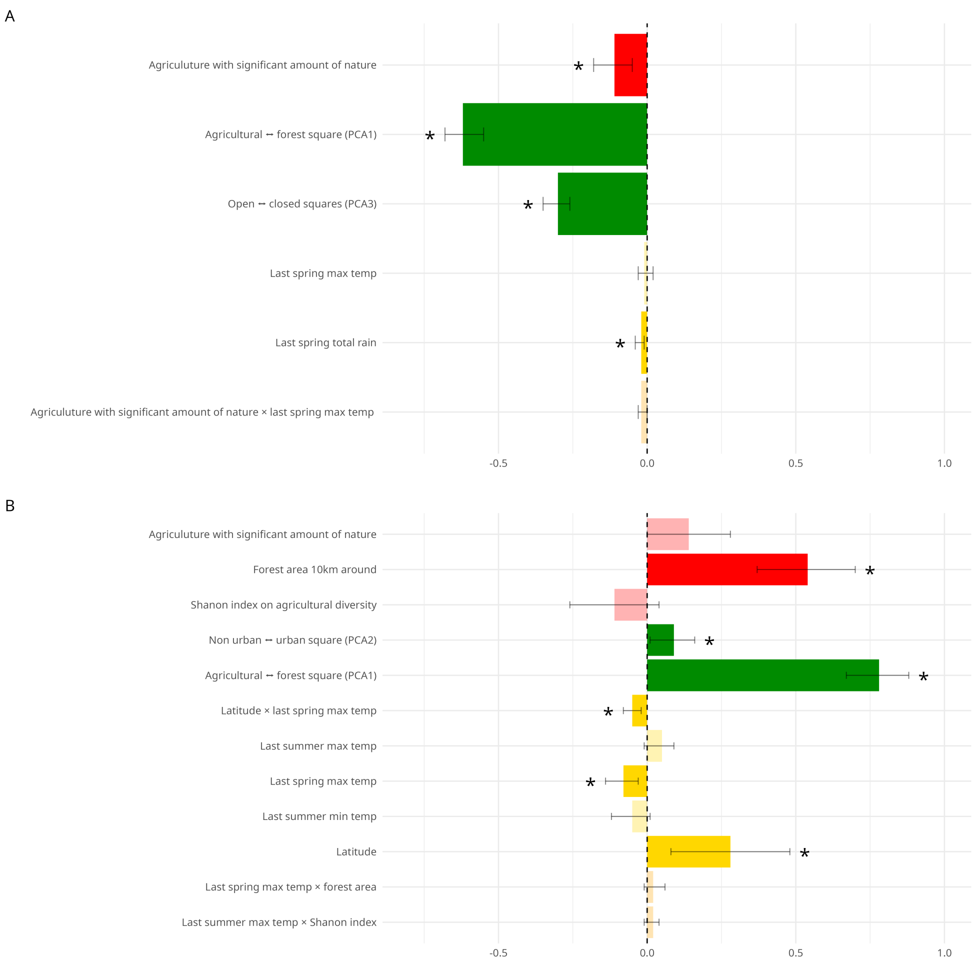

Fig. 3 Fixed effects of the best model for farmland species with conservation concern (A) and woodland species with conservation concern(B). The variables highlighted in red are the land sharing and land sparing variables, the ones in green are about the land use of the squares, the one in yellow represent climate variables and the orange one are interactions between land sharing and sparing variables and climate. Significant variables are represented in plain color and with a star whereas non-significant variables are transparent.

forest area, which aligns with our hypothesis. The variable "agriculture with a significant amount of nature" was also selected with a positive coefficient. This result can be explained by the study by Monnet et al. (2026) that showed pesticides impact all birds and the fewer constraint forest species face, the better they perform. Regarding **farmland species**, unexpectedly, our results showed that a land sparing scenario is also preferable. This may be explained by the fact that in Europe we are already mostly in a land sharing scenario (Karner et al. 2019) and that being further on the land sharing spectrum would imply a habitat that is not the one they are specialized in.

Our results showed that the land sparing scenario is better for **forest species with conservation concern** because the abundance increases with a greater forest area.

For **agricultural species with conservation concern** it may appear counterintuitive that their highest abundances are observed in land sparing scenarios, *i.e.*, areas with a low proportion of natural habitats such as bushes and hedges. This pattern might suggest that farmland birds of conservation concern do not benefit from these semi-natural elements. However, this interpretation is likely misleading. We argue that ground-nesting farmland bird species, which are now the primary focus of conservation efforts, have become particularly vulnerable in intensive agricultural landscapes. This vulnerability stems from decades of agricultural mechanization and intensification. Our analysis of land use within survey squares further supports this hypothesis because birds with conservation concern are more abundant in farmland and open landscapes. Hence, as shown in recent literature (Monnet et al. 2026; Rigal et al. 2023), farmland birds are declining because of the intensification of agricultural practices and pesticide use. Consequently, for conservation of these species the agricultural policies should encourage a scenario with large open fields but with a low-yield agriculture (Perrot et al. 2025).

In order to answer the question of how to allocate land use to conserve biodiversity our results showed that there is not one agricultural practice to prefer to conserve all groups of birds. Indeed, land sharing is better for generalist species whereas land sparing is better for forest and agricultural species. However, as shown in Rigal et al. (2023) farmland birds are declining; therefore, even if land sharing is not the primary solution, we still consider a more biodiversity friendly agriculture to conserve bird populations. As shown in Monnet et al. (2026) pesticides are one of the drivers of this decline, so reducing their use could therefore be a key solution. Overall, we cannot conclude that farmland birds with conservation concern prefer land sparing. Despite their apparent specialization in intensive agricultural landscapes, our findings suggest that conserving these species should prioritize reducing chemical inputs and adopting wildlife-friendly practices within larger field structures, rather than intensifying management further.

A main limitation of our analyses is the long time series of the STOC database. Indeed, over this timescale there are no data on pesticides use, as the ones used in Monnet et al. (2026) only start in 2015 and the ones used in Perrot et al. (2025) start in 2008. There are also no data available on small natural structures inside (hedgerow monitoring started in 2022 (Preux 2025)) fields. The indicators we used are thus, the most appropriate variables available for this study period to account for land use heterogeneity (with the Shannon index on agriculture practices), the presence of natural structures in fields (agriculture with a significant amount of nature), as well as big conserved forest. These three variables summarize well different characteristics Then, we did not take into account detection bias, but we aggregated observations at the survey-square scale to reduce randomness of detectability bias (Nabias et al. 2024).

In conclusion, our findings underscore a widespread challenge in agricultural landscapes akin to those in France: striking a balance between biodiversity conservation and food security, particularly for species of conservation concern that depend on habitats shaped by intensive farming.

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Declarations

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Competing interest

The authors have no relevant financial or non-financial interests to disclose.

Author contributions

All authors contributed to the study conception and design. Material preparation and (online) data collection was performed by Adélie Erard. Data analysis was performed by Adélie Erard and Romain Lorrillière. The first draft of the manuscript was written by Adélie Erard and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability

The bird abundance data analyzed during the current study are available on reasonable request on Vigie-Nature website. Climate data are available on WordClim website. Land use data are available on Copernicus Website.

Code availability

`R` code used for analysis is available at https://github.com/adelieerard/birds_inla.

Land Sharing or Land Sparing? Implications for Farmland Birds of Conservation Concern in France.

Supplementary material

Landscape Ecology

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List of Figures

List of Tables

1 Maps

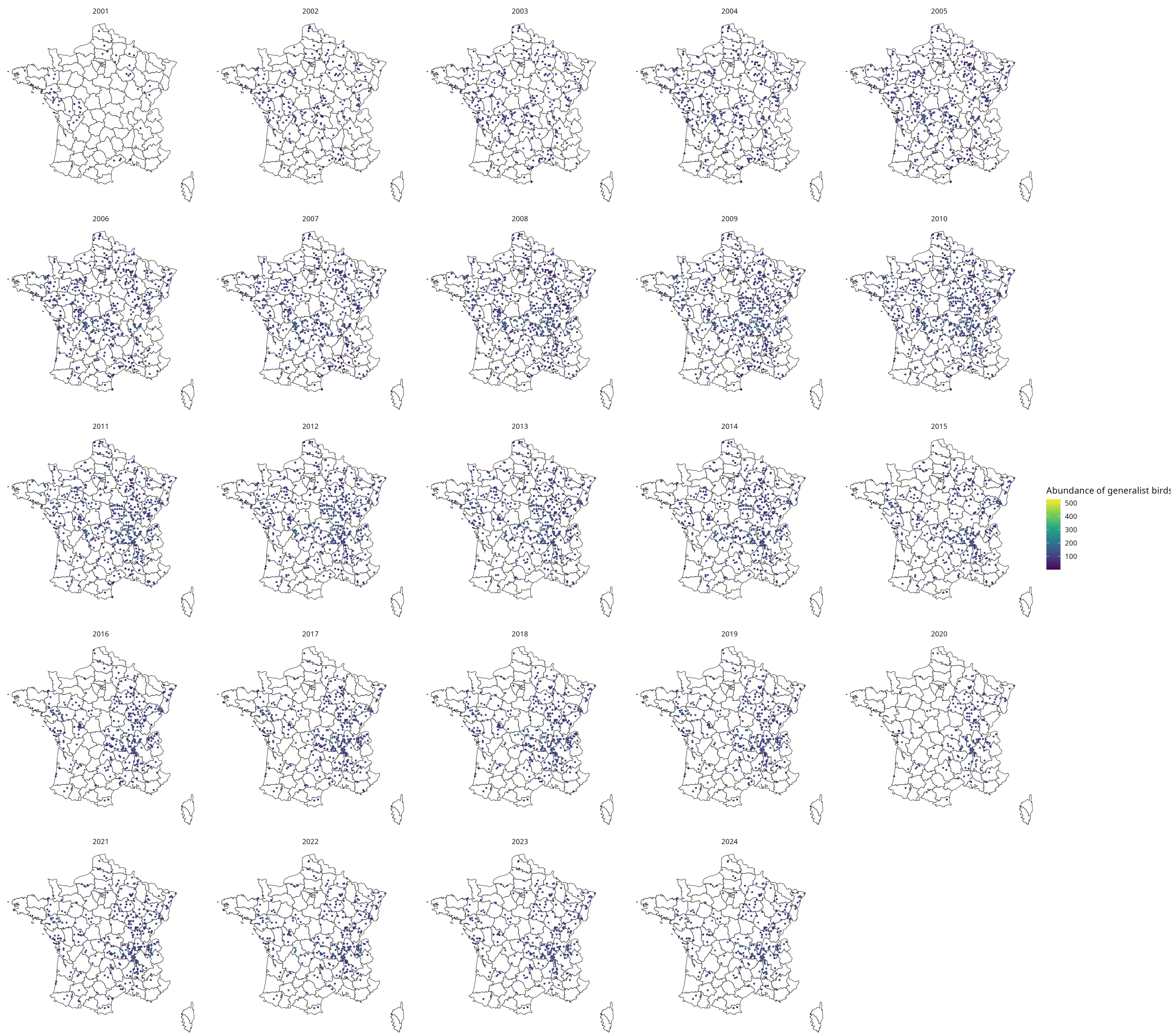


Figure SI-1: Observed abundance of generalist species, by year

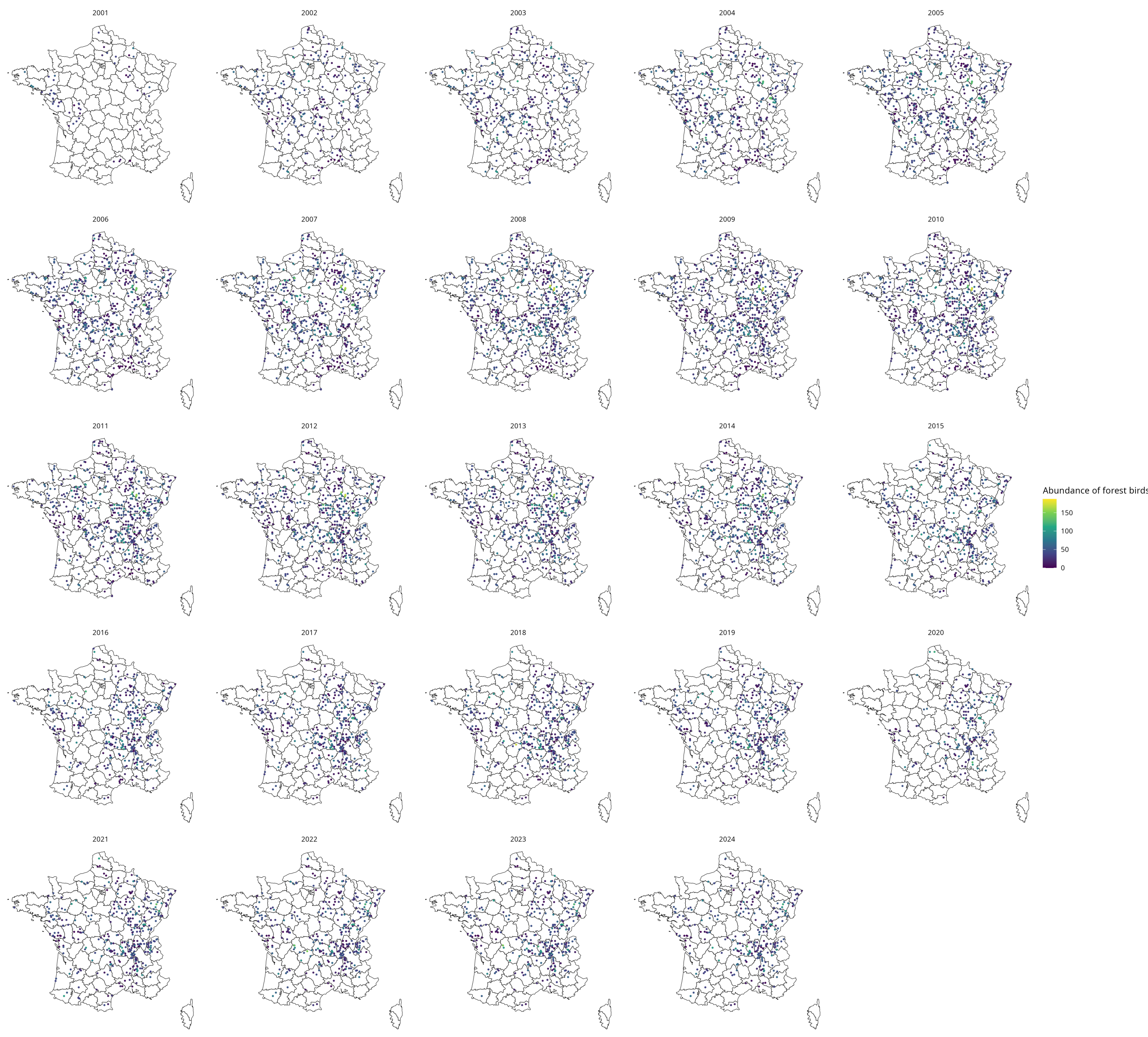


Figure SI-2: Abundance of forest species observed in the STOC program, by year

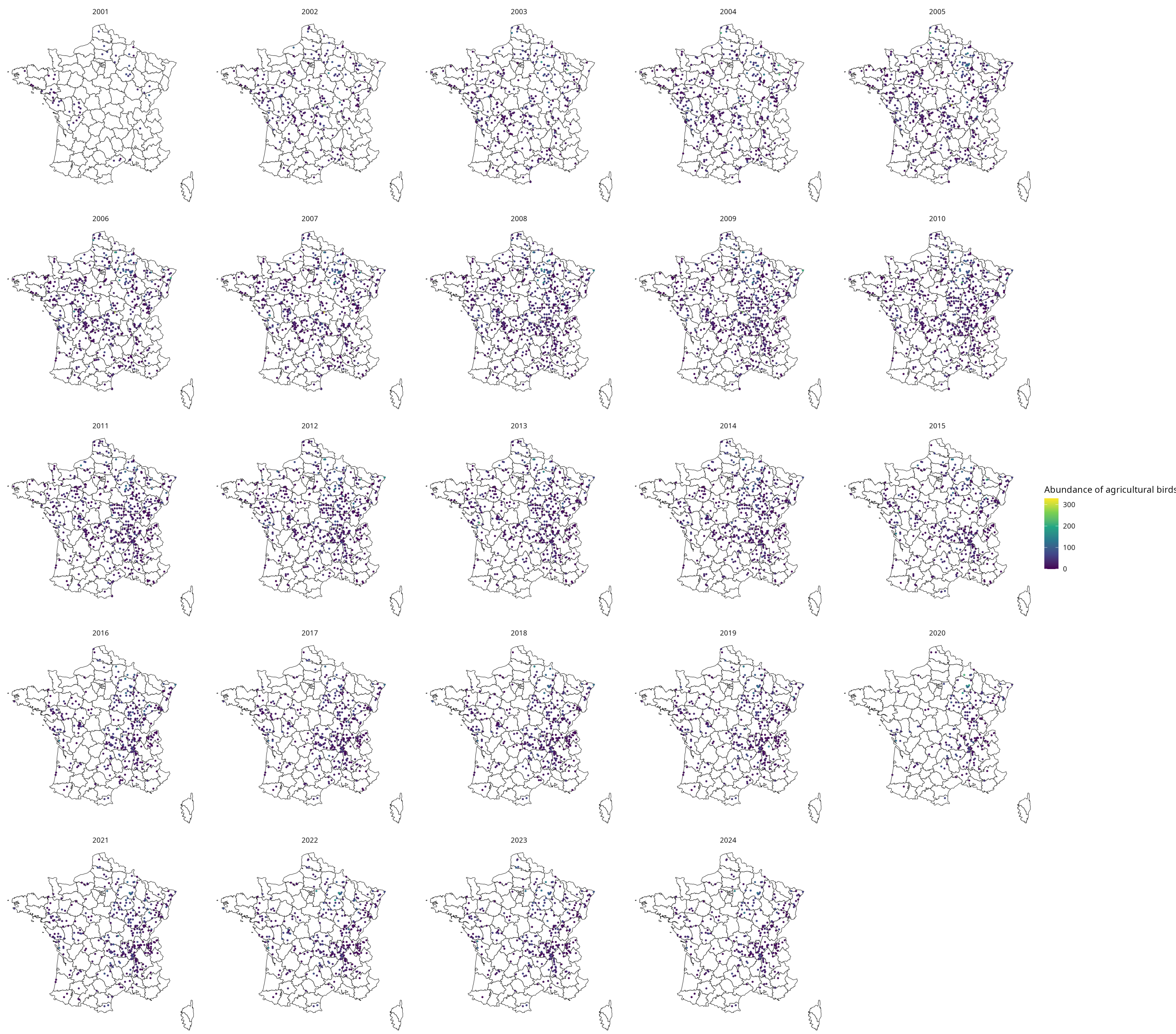


Figure SI-3: Observed abundance of agricultural species, by year

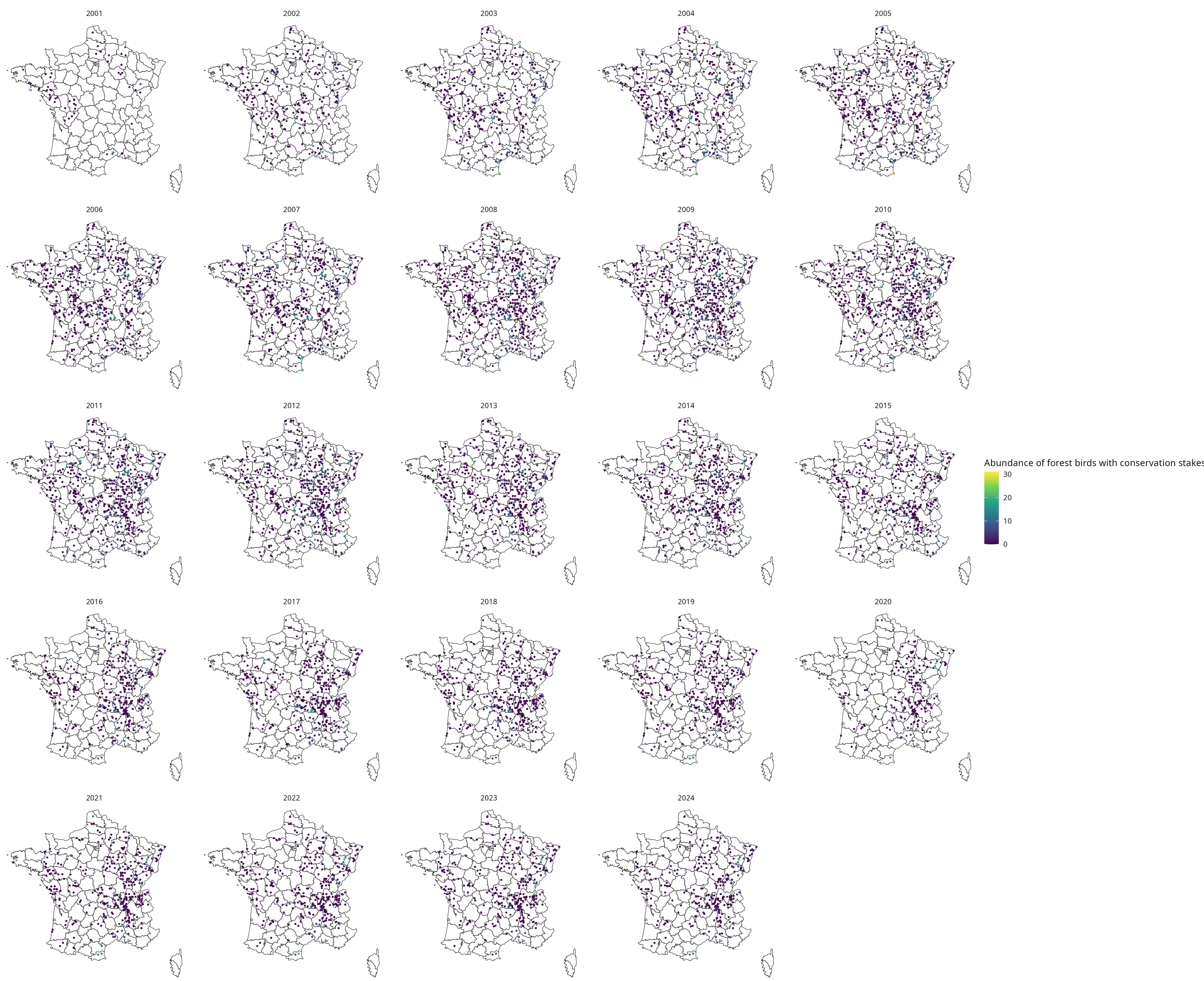


Figure SI-4: Observed abundance of forest species with conservation concern, by year

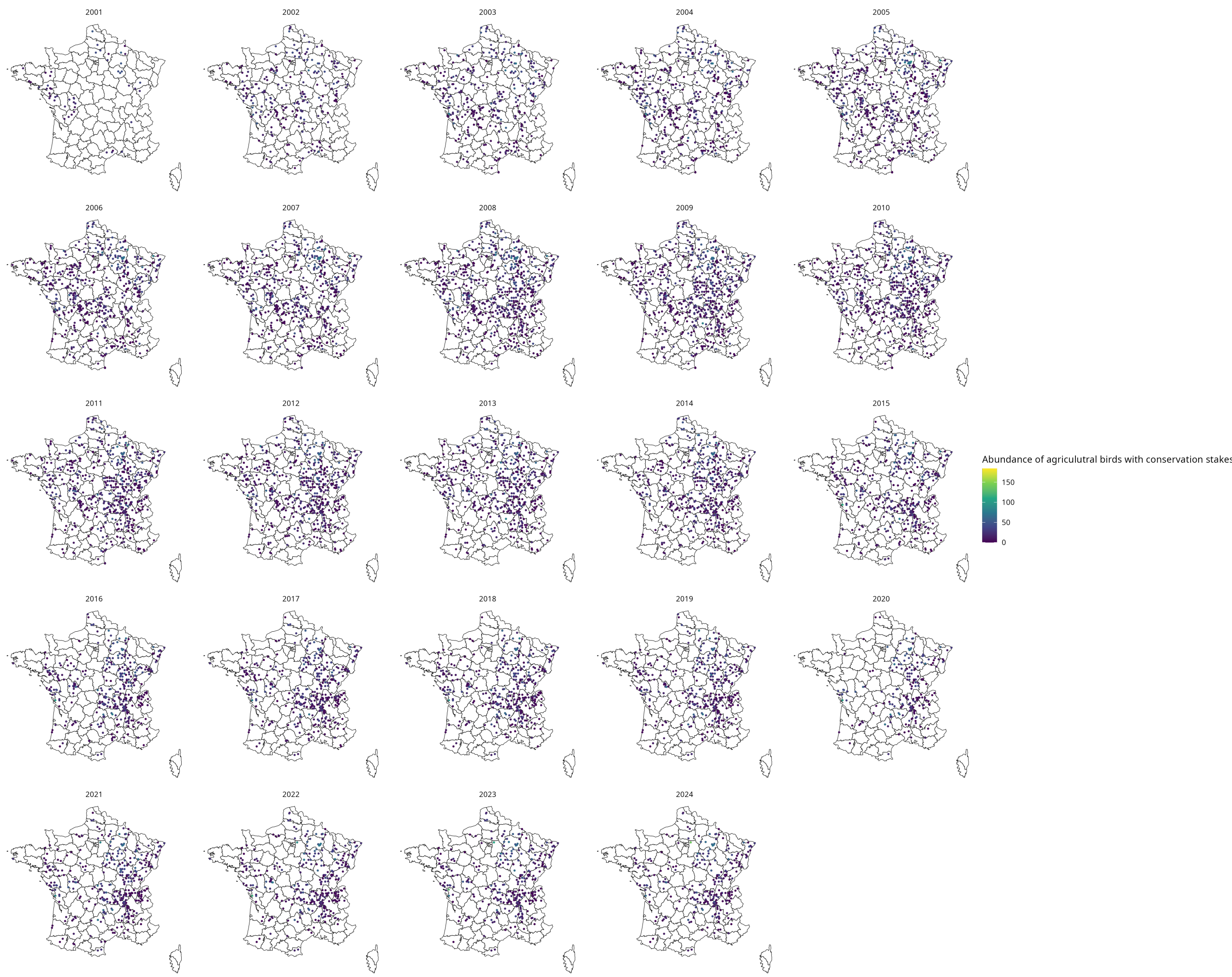


Figure SI-5: Observed abundance of agricultural species with conservation concern, by year

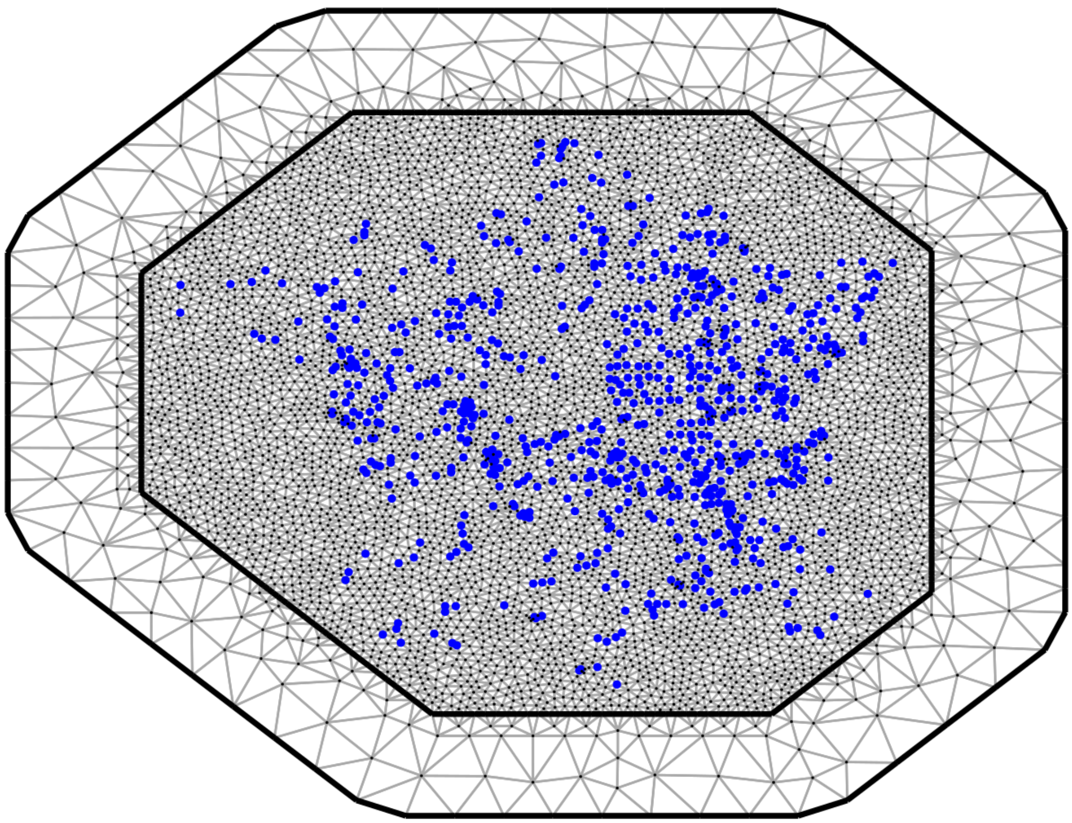
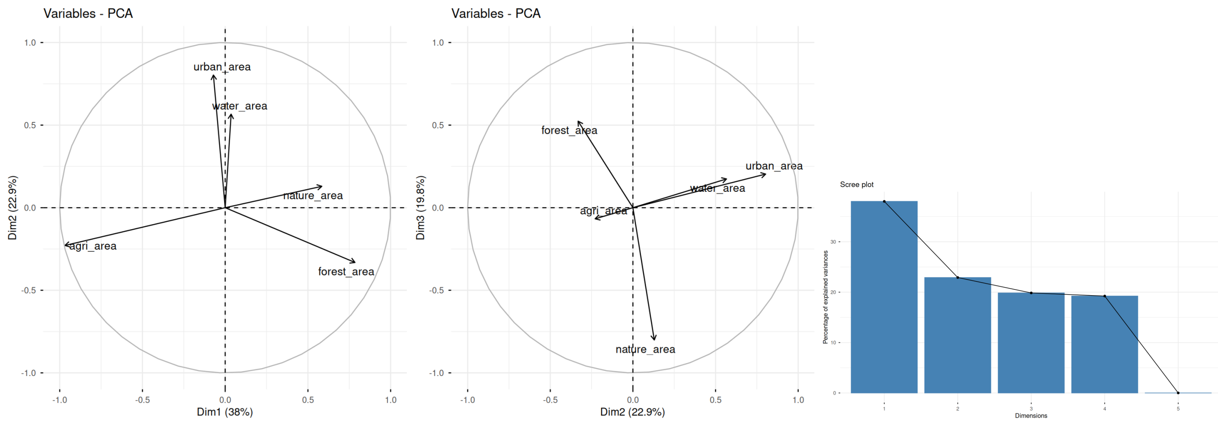


Figure SI-6: Mesh choice. The observed points are represented in blue.

2 PCA on land use of the square



(a) Axis 1 and 2 (b) Axis 2 and 3 (c) Eigenvalues

Figure SI-7: Results of the PCA on the land cover of each square

3 Tables of the results

	mean	sd	0.025quant	0.5quant	0.975quant	mode	kld
(Intercept)	1.45	0.05	1.36	1.45	1.55	1.45	0.00
Latitude	0.06	0.04	-0.01	0.06	0.13	0.06	0.00
Last spring max temperature	0.02	0.01	-0.00	0.02	0.04	0.02	0.00
Last summer max temperature	0.01	0.01	-0.01	0.01	0.03	0.01	0.00
PCA1	-0.52	0.03	-0.58	-0.52	-0.46	-0.52	0.00
PCA3	-0.23	0.02	-0.27	-0.23	-0.18	-0.23	0.00
Forest 10km around	-0.03	0.04	-0.11	-0.03	0.04	-0.03	0.00
Agriculture with signif nature	-0.05	0.03	-0.11	-0.05	0.01	-0.05	0.00
Last spring max temperature×Forest 10km around	0.01	0.01	-0.00	0.01	0.03	0.01	0.00
Last summer max temperature×Agriculture with signif nature	0.01	0.01	-0.00	0.01	0.02	0.01	0.00
Latitude×Last spring max temperature	0.02	0.01	0.01	0.02	0.03	0.02	0.00

	mean	sd	0.025quant	0.5quant	0.975quant	mode
(1/overdispersion)	11.64	0.28	11.09	11.64	12.21	11.64
Range for spatial (m)	17435.26	1365.42	14914.11	17377.90	20286.97	17255.39
Stdev for spatial	0.79	0.02	0.74	0.79	0.84	0.79
Precision for year	111.48	72.08	20.67	94.76	290.94	58.91
Rho for year	0.89	0.08	0.70	0.91	0.98	0.95

Table SI-1: Results for agricultural species

	mean	sd	0.025quant	0.5quant	0.975quant	mode	kld
(Intercept)	1.70	0.02	1.66	1.70	1.75	1.70	0.00
PCA1	0.44	0.02	0.40	0.44	0.49	0.44	0.00
PCA2	-0.07	0.02	-0.10	-0.07	-0.04	-0.07	0.00
Agriculture with signif nature	0.13	0.02	0.08	0.13	0.18	0.13	0.00
Latitude	0.23	0.03	0.17	0.23	0.30	0.23	0.00
Forest 10km around	0.17	0.03	0.11	0.17	0.23	0.17	0.00
Last spring max temperature	-0.00	0.01	-0.02	-0.00	0.02	-0.00	0.00
Last summer max temperature	-0.01	0.01	-0.03	-0.01	0.01	-0.01	0.00
Last spring min temperature	0.03	0.01	0.01	0.03	0.04	0.03	0.00
Forest 10km around×Last spring max temperature	-0.01	0.00	-0.02	-0.01	-0.01	-0.01	0.00
Forest 10km around×Last summer max temperature	0.01	0.00	0.01	0.01	0.02	0.01	0.00

	mean	sd	0.025quant	0.5quant	0.975quant	mode
(1/overdispersion)	47.25	1.71	43.99	47.21	50.71	47.13
Range for spatial (m)	18336.21	1369.44	15796.14	18282.59	21185.06	18170.46
Stdev for spatial	0.72	0.02	0.68	0.72	0.76	0.72
Precision for year	367.41	156.77	132.08	343.02	735.57	294.44
Rho for year	0.60	0.16	0.26	0.62	0.86	0.65

Table SI-2: Results for forest species

	mean	sd	0.025quant	0.5quant	0.975quant	mode	kld
(Intercept)	2.25	0.03	2.19	2.25	2.30	2.26	0.00
PCA1	0.03	0.01	0.01	0.03	0.06	0.03	0.00
Shanon index on agricultural patches	0.15	0.01	0.12	0.15	0.17	0.15	0.00
Latitude	0.05	0.01	0.02	0.05	0.08	0.05	0.00
PCA3	0.03	0.01	0.01	0.03	0.05	0.03	0.00

	mean	sd	0.025quant	0.5quant	0.975quant	mode
(1/overdispersion)	42.30	0.99	40.40	42.29	44.30	42.24
Range for spatial (m)	13071.93	1065.66	11080.66	13035.25	15273.40	12975.89
Stdev for spatial	0.39	0.01	0.36	0.39	0.42	0.39
Precision for year	359.25	243.30	67.33	300.15	980.45	185.71
Rho for year	0.89	0.08	0.69	0.91	0.98	0.95

Table SI-3: Results for generalist species

	mean	sd	0.025quant	0.5quant	0.975quant	mode	kld
(Intercept)	1.10	0.05	1.00	1.10	1.20	1.10	0.00
PCA3	-0.30	0.02	-0.35	-0.30	-0.26	-0.30	0.00
Last spring total rain	-0.02	0.01	-0.04	-0.02	-0.01	-0.02	0.00
Agriculture with signif nature	-0.11	0.03	-0.18	-0.11	-0.05	-0.11	0.00
Last spring max temperature	-0.01	0.01	-0.03	-0.01	0.02	-0.01	0.00
PCA1	-0.62	0.03	-0.68	-0.62	-0.55	-0.62	0.00
Agriculture with signif nature×Last spring max temperature	-0.02	0.01	-0.03	-0.02	-0.00	-0.02	0.00

	mean	sd	0.025quant	0.5quant	0.975quant	mode
(1/overdispersion)	29.51	1.37	26.92	29.47	32.31	29.38
Range for spatial (m)	18480.75	1459.90	15740.59	18434.65	21484.56	18366.57
Stdev for spatial	1.01	0.03	0.95	1.01	1.07	1.01
Precision for year	129.18	76.50	25.76	113.24	312.73	74.57
Rho for year	0.92	0.05	0.80	0.93	0.99	0.96

Table SI-4: Results for agricultural species with conservation concern

	mean	sd	0.025quant	0.5quant	0.975quant	mode	kld
(Intercept)	-0.48	0.12	-0.72	-0.48	-0.25	-0.48	0.00
Latitude	0.28	0.10	0.08	0.28	0.48	0.28	0.00
Last summer min temperature	-0.05	0.03	-0.12	-0.05	0.01	-0.05	0.00
Last spring max temperature	-0.08	0.03	-0.14	-0.08	-0.03	-0.08	0.00
Last summer max temperature	0.05	0.02	-0.00	0.05	0.09	0.05	0.00
PCA1	0.78	0.05	0.67	0.78	0.88	0.78	0.00
PCA2	0.08	0.04	0.01	0.08	0.16	0.08	0.00
Shanon index on agricultural patches	-0.11	0.08	-0.26	-0.11	0.04	-0.11	0.00
Forest 10km around	0.54	0.08	0.37	0.54	0.70	0.54	0.00
Agriculture with signif nature	0.14	0.07	-0.00	0.14	0.28	0.14	0.00
Last summer max temperature ×Shanon index on agricultural patches	0.02	0.01	-0.01	0.02	0.04	0.02	0.00
Last spring max temperature×Forest 10km around	0.02	0.02	-0.01	0.02	0.06	0.02	0.00
Latitude×Last spring max temperature	-0.05	0.02	-0.08	-0.05	-0.02	-0.05	0.00

	mean	sd	0.025quant	0.5quant	0.975quant	mode
(1/overdispersion)	12.87	1.47	10.24	12.78	16.00	12.60
Range for spatial (m)	36334.33	3889.39	29280.72	36127.94	44576.32	35719.07
Stdev for spatial	1.46	0.07	1.33	1.46	1.60	1.46
Precision for year	24.74	15.83	5.49	21.02	65.21	14.18
Rho for year	0.92	0.06	0.76	0.93	0.98	0.96

Table SI-5: Results for forest species with conservation concern

4 List of species with their habitat and conservation status

Table SI-6: Bird species by taxonomic order, habitat type and conservation status

Order	Species	Habitat type	Status
Accipitriformes	Buteo buteo	Agricultural	Least concerned
Passeriformes	Emberiza cirlus	Agricultural	Least concerned
Falconiformes	Falco tinnunculus	Agricultural	Conservation stake
Passeriformes	Lanius collurio	Agricultural	Conservation stake
Passeriformes	Emberiza citrinella	Agricultural	Conservation stake
Passeriformes	Corvus frugilegus	Agricultural	Least concerned
Passeriformes	Sylvia communis	Agricultural	Conservation stake
Charadriiformes	Vanellus vanellus	Agricultural	Conservation stake
Galliformes	Coturnix coturnix	Agricultural	Least concerned
Galliformes	Alectoris rufa	Agricultural	Least concerned
Upupiformes	Upupa epops	Agricultural	Least concerned
Passeriformes	Saxicola rubicola	Agricultural	Conservation stake
Passeriformes	Passer montanus	Agricultural	Conservation stake
Charadriiformes	Burhinus oedicnemus	Agricultural	Conservation stake
Passeriformes	Lullula arborea	Agricultural	Least concerned
Passeriformes	Alauda arvensis	Agricultural	Conservation stake
Passeriformes	Emberiza calandra	Agricultural	Conservation stake
Passeriformes	Carduelis cannabina	Agricultural	Least concerned
Galliformes	Perdix perdix	Agricultural	Least concerned
Passeriformes	Motacilla flava	Agricultural	Least concerned
Passeriformes	Anthus pratensis	Agricultural	Conservation stake
Passeriformes	Phylloscopus collybita	Forest	Least concerned
Passeriformes	Turdus philomelos	Forest	Least concerned
Piciformes	Dendrocopos major	Forest	Least concerned
Passeriformes	Erithacus rubecula	Forest	Least concerned
Passeriformes	Regulus ignicapilla	Forest	Least concerned
Passeriformes	Troglodytes troglodytes	Forest	Least concerned
Passeriformes	Certhia familiaris	Forest	Least concerned

Order	Species	Habitat type	Status
Passeriformes	Regulus regulus	Forest	Conservation stake
Passeriformes	Sitta europaea	Forest	Least concerned
Passeriformes	Lophophanes cristatus	Forest	Least concerned
Passeriformes	Phylloscopus bonelli	Forest	Least concerned
Passeriformes	Poecile palustris	Forest	Least concerned
Passeriformes	Certhia brachydactyla	Forest	Least concerned
Passeriformes	Turdus viscivorus	Forest	Least concerned
Passeriformes	Phylloscopus sibilatrix	Forest	Conservation stake
Passeriformes	Phylloscopus trochilus	Forest	Conservation stake
Passeriformes	Pyrrhula pyrrhula	Forest	Conservation stake
Passeriformes	Coccothraustes coccothraustes	Forest	Least concerned
Piciformes	Dendrocopos medius	Forest	Least concerned
Passeriformes	Parus montanus	Forest	Least concerned
Passeriformes	Sylvia melanocephala	Forest	Conservation stake
Columbiformes	Columba palumbus	Generalist	Least concerned
Passeriformes	Corvus corone	Generalist	Least concerned
Passeriformes	Cyanistes caeruleus	Generalist	Least concerned
Passeriformes	Sylvia atricapilla	Generalist	Least concerned
Passeriformes	Turdus merula	Generalist	Least concerned
Passeriformes	Fringilla coelebs	Generalist	Least concerned
Passeriformes	Hippolais polyglotta	Generalist	Least concerned
Passeriformes	Parus major	Generalist	Least concerned
Passeriformes	Garrulus glandarius	Generalist	Least concerned
Cuculiformes	Cuculus canorus	Generalist	Least concerned
Passeriformes	Oriolus oriolus	Generalist	Least concerned
Piciformes	Picus viridis	Generalist	Least concerned
Passeriformes	Luscinia megarhynchos	Generalist	Least concerned
Passeriformes	Prunella modularis	Generalist	Least concerned

5 Corine Land cover Codes

CLC Code	Class name
111	Continuous urban fabric
112	Discontinuous urban fabric
121	Industrial or commercial units
122	Road and rail networks and associated land
123	Port areas
124	Airports
131	Mineral extraction sites
132	Dump sites
133	Construction sites
141	Green urban areas
142	Sport and leisure facilities
211	Non-irrigated arable land
212	Permanently irrigated land
213	Rice fields
221	Vineyards
222	Fruit trees and berry plantations
223	Olive groves
231	Pastures
241	Annual crops associated with permanent crops
242	Complex cultivation patterns
243	Land principally occupied by agriculture, with significant areas of natural vegetation
244	Agroforestry areas
311	Broad-leaved forest
312	Coniferous forest
313	Mixed forest
321	Natural grasslands
322	Moors and heath land
323	Sclerophyllous vegetation
324	Transitional woodland-shrub
331	Beaches, dunes, sands

332	Bare rocks
333	Sparsely vegetated areas
334	Burnt areas
335	Glaciers and perpetual snow
411	Inland marshes
412	Peat bogs
421	Salt marshes
422	Salines
423	Intertidal flats
511	Water courses
512	Water bodies
521	Coastal lagoons
522	Estuaries
523	Sea and ocean

Table SI-7: Corine Land Cover (CLC) Level 3 classification codes and class names